



CANDIDATE NAME

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CT GROUP

17S7 _____

CENTRE NUMBER

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**INDEX
NUMBER**

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

14 September 2018

No Additional Materials are required.

2 hours

INSTRUCTIONS TO CANDIDATES

There are **four** question booklets (**I to IV**) to this paper. Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page, and your **name** and **CT group** on the lines provided at the top of the cover page of Booklets **II, III and IV**.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 25
2	/ 13
3	/ 12
4 or 5	/ 25
Final Mark	/ 75

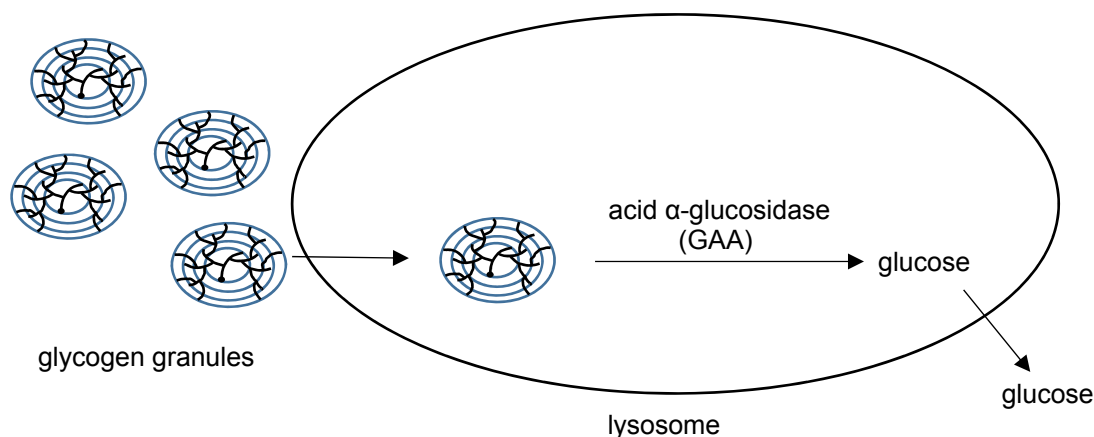
This document consists of **20** printed pages.

BOOKLET I

SECTION A

QUESTION 1

Glycogen is a readily mobilised storage form of glucose. Fig. 1.1 shows how glycogen metabolism occurs within smooth muscle cells when energy is required. A lysosomal enzyme, acid α -glucosidase (GAA), catalyses this reaction. GAA is the only enzyme that hydrolyses glycogen to glucose in the acidic environment of the lysosome.

**Fig 1.1**

(a)(i) Describe how the structure of glycogen is related to its function.

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(ii) Explain the significance of localising GAA in the lysosomes.

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Autophagy involves degradation and recycling of both damaged and worn out organelles such as mitochondria and lysosomes. Lysosomes are key players in the autophagic process. Impaired autophagy is associated with a wide range of diseases such as Pompe disease.

Pompe disease is a lysosomal glycogen storage disease caused by deficiency of GAA. The deficiency of GAA leads to glycogen accumulation within the lysosome and subsequent enlargement of the organelle. Cells with enlarged lysosomes are damaged and undergo extensive autophagy which may be unsuccessful.

To determine whether an individual suffers from Pompe disease, smooth muscle cells are isolated and observed. Cells were collected from six healthy and six diseased individuals each. These cells were categorised based on the amount and localisation of glycogen deposits in the cells as follows:

- types 1 to 3 showed varying degrees of glycogen deposits in the cells, but not specifically within lysosomes
 - type 1: no visible glycogen granules
 - type 2: scattered glycogen granules
 - type 3: large amount of glycogen granules
- type 4 cells showed significant glycogen deposits within lysosomes
- type 5 cells showed extensive autophagy of glycogen-containing lysosomes.

Fig. 1.2 shows the frequency of cells with the varying types of glycogen deposits found in both healthy and diseased individuals.

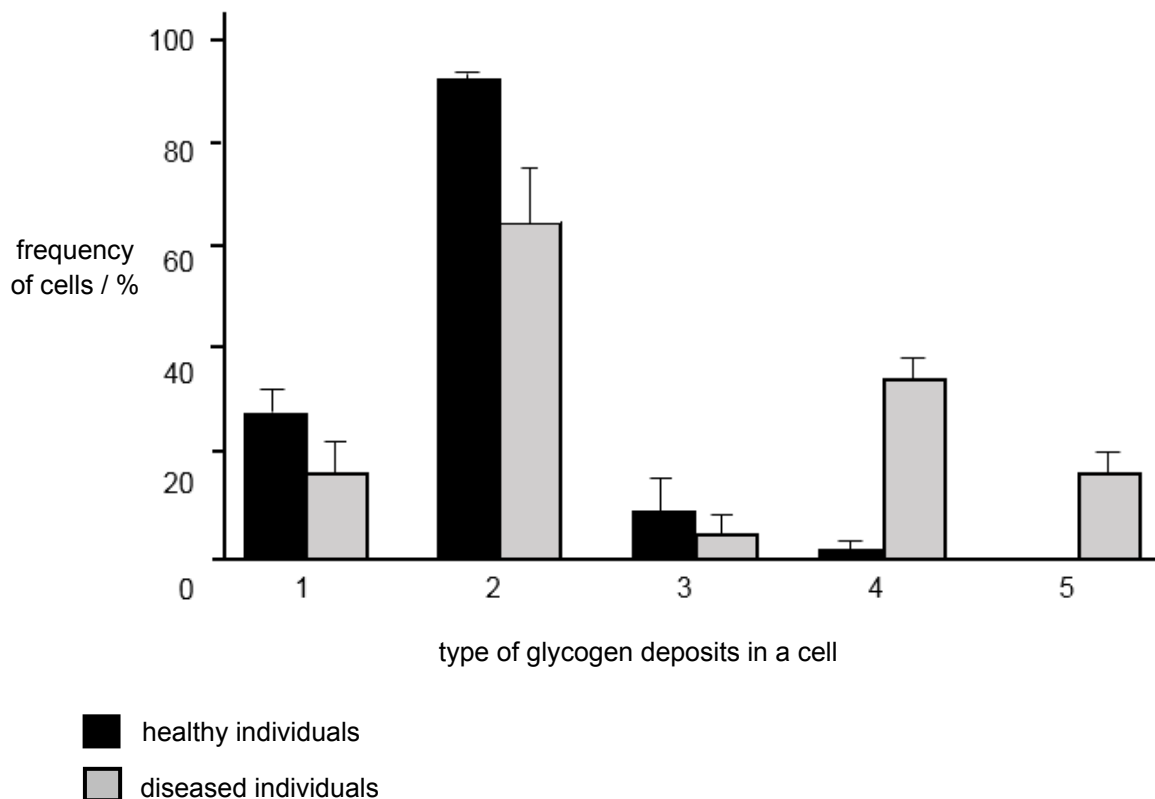


Fig. 1.2

(b)(i) Account for the results shown in Fig. 1.2.

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(ii) Suggest how the mutated *GAA* gene results in cellular damage.

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Pompe disease is inherited in an autosomal recessive manner. Fig. 1.3 shows the pedigree analysis of two families with more than one individual affected with Pompe disease.

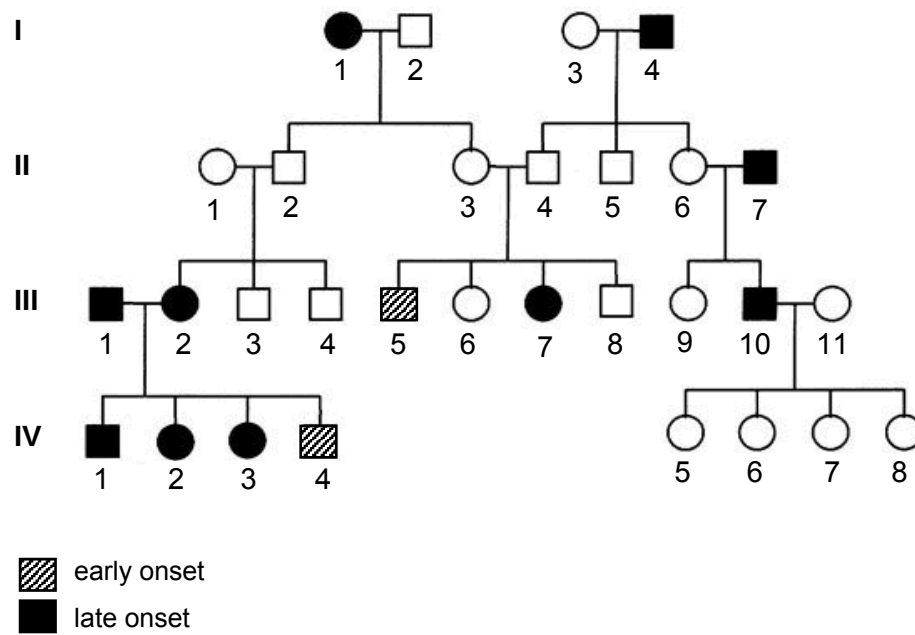


Fig. 1.3

(c)(i) With reference to Fig. 1.3, explain why Pompe disease is an autosomal recessive disease.

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(ii) None of the children of individuals **III-1** and **III-2**, other than individual **IV-4**, suffered from early onset Pompe disease. Suggest in whom and in what organ a mutation occurred that accounts for this.

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In Pompe disease, there is a broad spectrum of phenotypes that range from infantile-onset form to late-onset form. Disease severity primarily depends on level of GAA enzymatic activity that is related to the different mutations in the GAA gene.

Table 1.1 shows the three types of classification for Pompe disease.

Table 1.1

	classic infantile-onset	non-classic infantile-onset	late-onset
onset	few months of birth	by the age of 1	later in childhood, adolescence, or adulthood
severity	most severe	moderately severe	least severe
symptoms	progressive muscle loss, loss of motor, respiratory and cardiac functions leading to respiratory failure		

(d) Explain how genotype is linked to phenotypes of Pompe disease.

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[3]

Enzyme replacement therapy (ERT) is an approved treatment for Pompe individuals. Active GAA enzymes are extracted and purified from *in vitro* sources. During ERT, these enzymes are administered to the Pompe individuals through intravenous infusions directly into the bloodstream on a weekly basis.

(e)(i) Explain how ERT is used to treat Pompe individuals.

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(ii) Suggest why Pompe individuals require weekly infusions of active GAA enzymes.

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[Total: 25]

QUESTION 2

Almost a decade before the discovery of penicillin, the controversial practice of bacteriophage therapy was being developed as a treatment for bacterial infections in humans. However, it was eventually eclipsed by the rise of antibiotics.

Research into bacteriophage therapy is increasing in popularity in recent years due to the emergence of pathogenic bacteria resistant to most, if not all, currently available antibiotics. This has become a critical problem in modern medicine.

Fig. 2.1 shows how bacteriophage therapy was carried out.

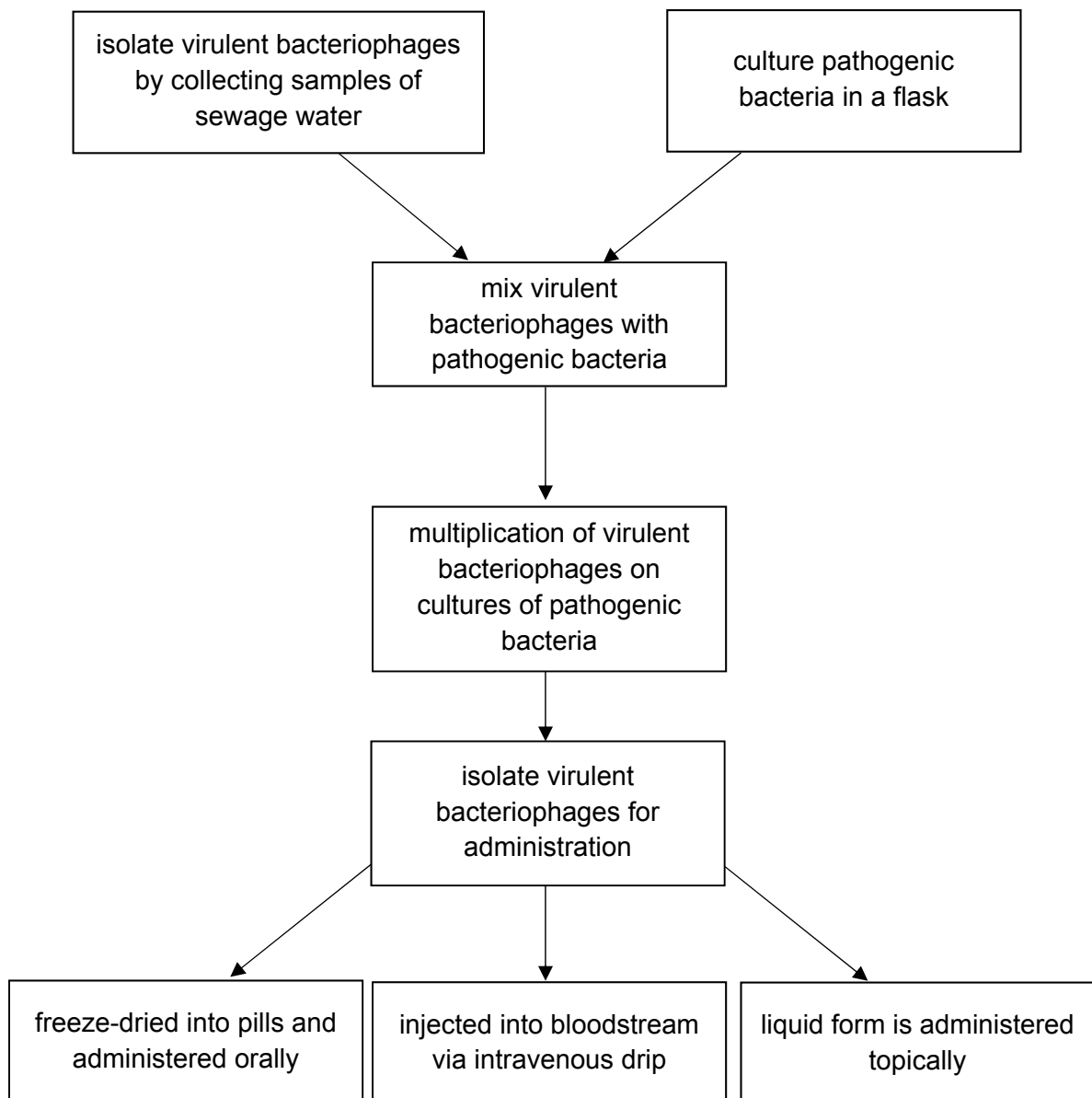


Fig. 2.1

- (a) Explain why there is a need to mix isolated bacteriophages with pathogenic bacteria.

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- (b) With reference to its reproductive cycle, suggest how a virulent bacteriophage could be used to treat bacterial infections.

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- (c) Explain why bacteriophage therapy may be a better mode of treatment than antibiotics.

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- (d) Suggest and explain for the controversial nature of this form of bacteriophage therapy.

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[Total: 13]

QUESTION 3

Bacteria may be classified according to differences in cell wall structure. The differences are shown by using the Gram stain.

- A Gram-positive bacterium has a cell wall mainly composed of a thick layer of peptidoglycan.
- A Gram-negative bacterium has a more complex cell wall. This wall is composed of a much thinner layer of peptidoglycan and an outer layer known as the outer membrane.

Escherichia coli is a Gram-negative bacterium.

Fig. 3.1 is a diagram through the cell surface membrane and the cell wall of *E. coli*.

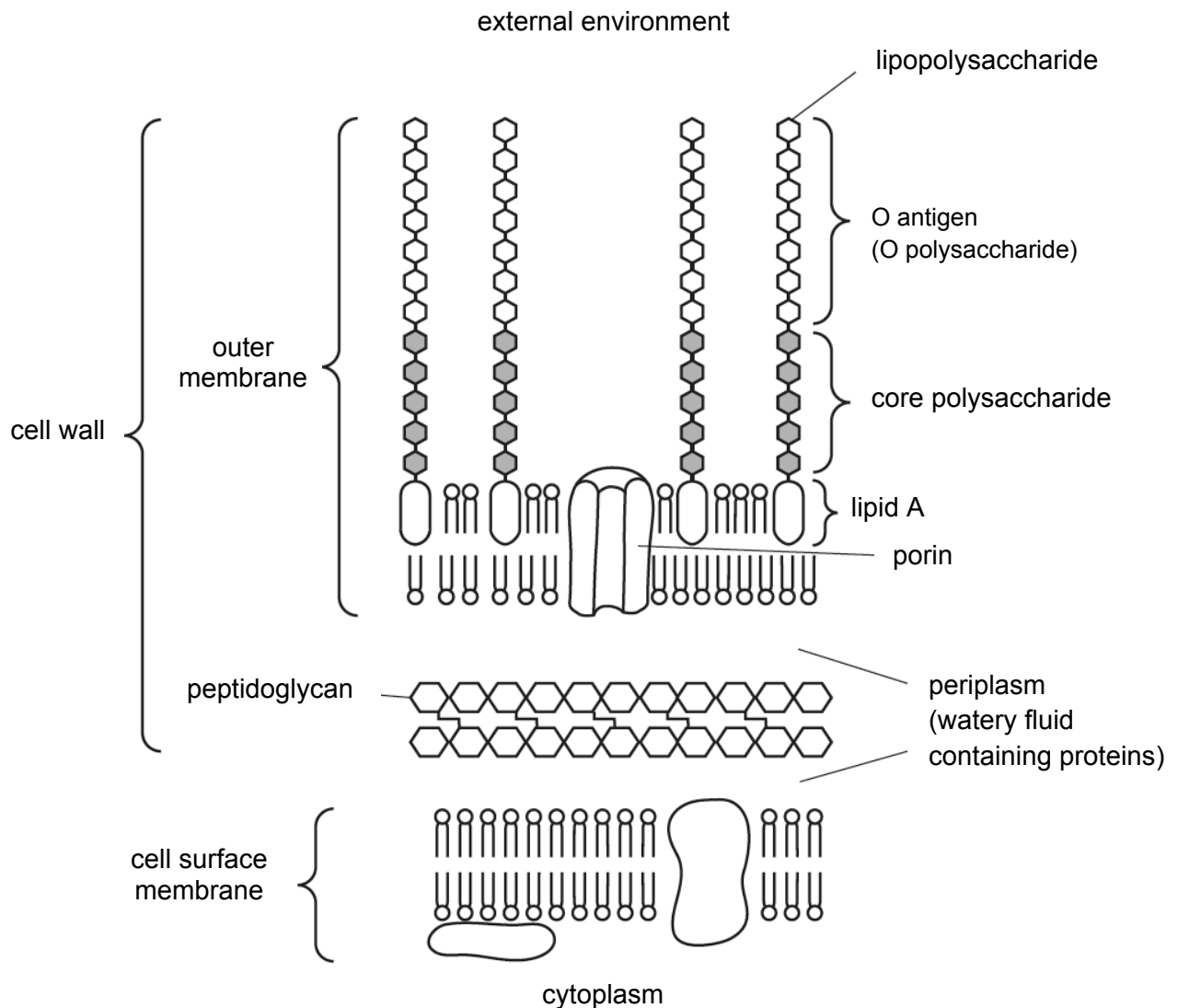


Fig. 3.1

- (a) The antibiotic penicillin kills bacteria by causing them to lyse. It is more effective in treating diseases caused by Gram-positive bacteria than diseases caused by Gram-negative bacteria.

Outline how penicillin acts on bacteria and use Fig. 3.1 to suggest why penicillin has little or no effect at treating diseases caused by Gram-negative bacteria, such as some strains of *E. coli*.

[4]

- (b)** Using Fig. 3.1, suggest the possible roles of the periplasm in bacteria.

[2]

Fig. 3.1 shows that the outer membrane of the cell wall of *E. coli* contains lipopolysaccharides, which are not present in the cell surface membrane.

(c)(i) Define the term *lipopolysaccharide*.

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(ii) The O antigen is the outer part of the polysaccharide portion of the lipopolysaccharide, facing the aqueous external environment. Some strains of *E. coli* are pathogenic. Different pathogenic strains have different O antigens.

Suggest and explain why infection with one pathogenic strain of *E. coli* does not provide immunity to a different pathogenic strain.

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[Total: 12]

--- END OF SECTION A ---

SECTION B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

QUESTION 4

- (a)** Eukaryotic chromatin, the DNA-nucleosome polymer, is a dynamic molecule that exists in different configurations during the course of the cell cycle.

Describe how DNA is packaged in a cell and discuss the advantages of the eukaryotic chromatin being a dynamic molecule. [14]

- (b)** The study of identical twins provides an opportunity for exploring the extent to which epigenetics and environmental factors contribute to phenotypic variation in humans. Epigenetics is the study of heritable changes in gene expression that do not involve changes to the underlying DNA sequence. Epigenetic silencing turns gene off, contributing to differential gene expression in specialised cells.

Describe how epigenetic silencing may occur and discuss the extent to which epigenetic silencing accounts for identical twins being phenotypically non-identical. [11]

[Total: 25]

QUESTION 5

- (a)** The flow of electrons in oxidation-reduction reactions is responsible, directly or indirectly, for all work done by living organisms. Plants generate cellular energy in the form of ATP by the same process of chemiosmosis.

Describe the process of chemiosmosis and discuss the importance of electron flow in both chloroplasts and mitochondria that leads to chemiosmosis, with particular emphasis on the role of membrane proteins. [14]

- (b)** In plants, carbon dioxide (CO₂) and water (H₂O) are the products of cellular respiration while they are the reactants in photosynthesis. Tropical rain forests are estimated to be responsible for approximately 20% of global photosynthesis, yet the consumption of large amounts of CO₂ by living trees is thought to make little or no net contribution to reduction of global warming.

Explain how photosynthesis and cellular respiration are related to the climate and discuss the contributing factors that continue to bring about global warming even with large amounts of CO₂ being consumed by trees in the tropical rainforests. [11]

[Total: 25]

--- END OF SECTION B ---

--- END OF PAPER---

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